



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

COMPUTER SCIENCE

PAPER 2 Practical

6023/2

3 hours

SPECIMEN PAPER

Additional materials:

CD / flash disk

Printing facility

Bond paper

INSTRUCTIONS TO CANDIDATES

This is a purely practical examination. All answers should be printed. Handwritten answers will **not** be marked.

All work should be backed up by a soft copy on a CD/flash disk. Work on CD/flash without hard copy will not be marked.

Each answer sheet should include the following information in the header section:

- Candidate Name and Candidate Number
- Centre Name and Date
- Subject Code

Print screenshots for all designs.

Web design print outs must clearly show the address bar of the web page.

When answering programming questions, indicate the language used.

If you print on more than one sheet, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This specimen paper consists of 6 printed pages.

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Section A [20 marks]

Answer any **one** question from this section.

- 1 (a) Construct a website with two web pages namely Home and Application for your school. The two web pages should be linked by hyperlink. [4]

- (b) Use your school name as the main heading on the home page and sub heading named 'About us' and include the following paragraph.
Our school offers a variety of Subject areas at *O level* and *A level*. The school has adequate classrooms with state of the art equipment to provide a conducive environment for learning. School fees is now affordable, it has dropped from \$600 to \$550 and it should be payable by 13th of January.

Apply the following formats on text in the paragraph.

- Italicise the phrases 'O level' and 'A level'.
- Change font colour of phrase 'provide a conducive environment for learning' to blue.
- Apply strike through on '\$600'.
- Apply superscript on 'th' on the date. [4]

- (c) Create a form on the Application page.

Objects on the form include

- form title 'Advanced level Application form',
- labels and textboxes to capture Student Name, DOB and Phone number,
- radio buttons to capture Gender,
- check boxes to select desired subjects,
- apply button. [12]

- 2 A newly established hotel **Gokwe Villa** has approached you as a web developer. Construct a website for the hotel. The website has only one page.

Create a form to allow users to enter customer details and select a room.

The form must display

- customer details
- details of the room (executive, single or double)
- cost of the room per day
- the total cost

Users must be able to enter the start date and check out date.

Users must be able to select a room from the drop down list.

Users must not be able to edit other fields. [20]

Section B [50 marks]

Answer any **one** question from this section.

- 3 (a) Create a program for ECD learners. The program accepts a letter of alphabet from a learner. If the learner enters **a,e,i,o** or **u**, the program should display a message indicating that the letter is a vowel. If the learner enters **y**, the program should display a message indicating that sometimes **y** is a vowel and sometimes **y** is a consonant. Otherwise, the program should display a message indicating that the letter is a consonant. [15]

- (b) Produce program code that reads the content of File 1 and then writes it to File 2 as follows:

* use Consolas / Courier font

< Line 1>

< Line 2> < centre number >

< Line 3> < candidate number >

< Line 4> < Name, Surname >

< Date / time stamp >

Sample Output

This is my source file

Number 1 is: 0100015

Number 2 is: 0059

My Name is: Chipu Made

01/01/ 2020

* end of Consolas / courier font

NB: Submit only your source code for this question. [15]

- (c) Produce a program using the high level programming language that initialises an array of 20 characters with capital letters of the alphabet.

NB: ASCII code for 'A' = 65 [8]

- (d) Produce a program by copying and modifying your code from (d) above or otherwise.

The program in this case works with a 2 D array, each element containing two characters. (capital letters and small letters)

- Array should be a 20 × 20 elements.

NB: ASCII code for 'a' = 97. [12]

- 4 (a) The following section of pseudocode inputs 100 numbers and outputs the **largest positive** and **largest negative** number. There are some errors in the pseudocode.

1. Input number
2. neg = number: pos = number
3. For x = 0 to 100
4. Input number
5. If number > pos then number = pos
6. If number < neg then number = neg
7. x = x + 1
8. Next x
9. Output pos, neg

Correct the errors and develop a program for the pseudocode. [10]

- (b) Develop a program that processes invoices for a company selling a variety of products. It should prompt the user to enter the unit cost of the product, how many were sold and the date the invoice was supposed to be paid. Use two functions, the first one calculates and returns the basic cost of the invoice including VAT (15%) and the second function reduces the basic cost by 10% if the invoice has been paid on time. [20]
- (c) Develop a program for a particular university. Students on a course take three module exams. Students can have two attempts at each exam module and the best result counts. Declare a class called student that stores the student's ID, name and three exam marks (out of 100). If the best mark for a given module is the student's second attempt, store the second attempt mark, otherwise keep the first mark. The program should output how many modules the student passed. A pass mark is 50 and above. Determining the number of passes should be one of the class' methods. [20]

Section C [30 marks]

Answer any **one** question from this section.

- (a) Gabaza video club maintains a database of movies to be rented out. Without normalisation, all information is stored in one table as shown below.

Member ID	Title	Full Name	Address	Movie Rented	Category
1	Mrs	Jane Moyo	902 Mutare	Pirates Titans	Action, Action
2	Mr	Robert Dube	54 Gweru	Forgotten Daddy	Educational Comedy
3	Mrs	Anesu Ncube	100 Airport	Titans	Action

Normalise the above table to second normal form.

[14]

(b)

Student-ID	F Name	Surname	Sport	Venue
4241	Sibo	Nare	cricket	Harare
4741	Takunda	Mafu	soccer	Harare
4858	Farai	Muganu	volley	Kwekwe
4740	Sean	Gutsa	volley	Mutare
4848	Layla	Tanguni	cricket	Kwekwe
4742	Sindiso	Moyo	cricket	Mutare
4444	Thubani	Malahleka	soccer	Mutare

- (i) Using DBMS of your choice, identify the primary key and create the sport table above. [4]
- (ii) Create a query to display the records of students who play cricket, print the screen shot and the SQL code. [4]
- (iii) Produce the delete query to remove Farai Muganu from the table. Print the screen shot and SQL code. [4]
- (iv) Produce the updated query to add the following record: 4748, Kenneth, Shawa, soccer, Gweru. Print the result and the SQL. [4]

- 6 A school website has a relational database, SCHOOLWEB, that stores the website's information.
The Database has 3 tables to store student details, image details and text that they post.
The tables are as shown below:

Student (**Student ID**, Surname, First Name, Date of Birth)

Photo (**Photo ID**, Student ID, Comment, Upload Date)

Text Post (**Post ID**, Student ID, Date of post, The Text)

- (a) Draw the ERD to show the relationship between the 3 tables. For fields, only include the primary keys on the ERD. [9]
- (b) The following table shows sample data for a **student** table.

Student ID	First Name	Surname	Date of Birth
tinol23	Tinotenda	Mutamba	01/01/95
jackmoyo	Jacob	Moyo	23/02/56
muteroa	Anesu	Mutero	03/03/97
bigboss3	Mazvita	Tongogara	04/03/82

- (i) Produce an SQL statement to create the student table above. [5]
- (ii) Produce an SQL statement to add a new field, country, to the student table. [2]
- (iii) Produce an SQL statement to display all attributes for students who were born before 1990. [3]
- (iv) Produce an SQL statement to display all records in descending order of student surname. [3]
- (c) Draw a well-labelled diagram showing the 3-layered schema of the Database Management System (DBMS). [8]